Recovery of metal of interest from the complex by reduction:

Gold can be recovered by reacting the deoxygenated leached solution with zinc. In this process the gold is reduced to its elemental state (zero oxidation sate) and the process is called **cementation**.

$$Zn\ (s) + 2Au(CN)_2]^-\ (aq) \longrightarrow [Zn(CN)_4]_2^-(aq) + 2Au\ (s)$$

Ammonia leaching

When a crushed ore containing nickel, copper and cobalt is treated with aqueous ammonia under suitable pressure, ammonia selectively leaches these metals by forming their soluble complexes viz. $[Ni(NH_3)_6]^{2+}$, $[Cu(NH_3)_4]^{2+}$, and $[Co(NH_3)_5H_2O]^{3+}$ respectively from the ore leaving behind the gangue, iron(III) oxides/hydroxides and aluminosilicate.

Alkali leaching

In this method, the ore is treated with aqueous alkali to form a soluble complex. For example, bauxite, an important ore of aluminum is heated with a solution of sodium hydroxde or sodium carbonate in the temperature range 470 - 520 K at 35 atm to form soluble sodium meta-aluminate leaving behind the impurities, iron oxide and titanium oxide.

$$Al_2O_3\ (s) + 2NaOH\ (aq) + 3H_2O\ (l) \longrightarrow 2Na[Al(OH)_4]\ (aq)$$

The hot solution is decanted, cooled, and diluted. This solution is neutralised by passing CO_2 gas, to the form hydrated Al_2O_3 precipitate.

$$2Na[Al(OH)_4]\ (aq) + CO_2\ (g) \longrightarrow Al_2O_3.xH_2O\ (s) + 2NaHCO_3\ (aq)$$

The precipitate is filtered off and heated around 1670 K to get pure alumina Al_2O_3.

Acid leaching

Leaching of sulphide ores such as ZnS, PbS etc., can be done by treating them with hot aqueous sulphuric acid.

$$2ZnS\ (s) + 2H_2SO_4\ (aq) + O_2(g) \longrightarrow 2ZnSO_4\ (aq) + 2S\ (s) + H_2O$$

In this process the insoluble sulphide is converted into soluble sulphate and elemental sulphur.

Evaluate yourself 1

1. Write the equation for the extraction of silver by leaching with sodium cyanide and show that the leaching process is a redox reaction.

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